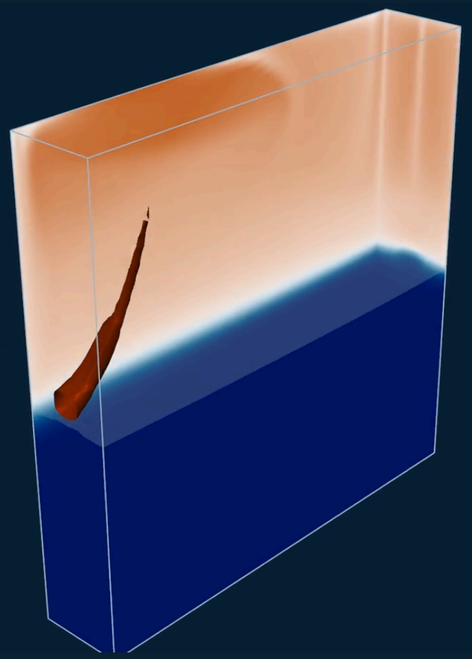
JOHANNES KEPLER
UNIVERSITY LINZ

RESEARCH TALK FOR AN INDUSTRIAL PARTNER

The flow settles in seconds. The temperature takes *hours*.

Record once, replay many times: data-based simulation of heat in a buoyant flow

Department of Particulate Flow Modelling
Johannes Kepler University Linz



cold plumes under a cooled lid, reference simulation, 3D render of stored solver states

THE QUESTION

Most of the cost goes into the *flow*, not into the temperature we want.

569 min

solver time for about ten minutes of flow in a 75 mm tank

Can we keep the temperature physics, and stop paying for the flow every time?

It depends on one thing: does the temperature change the flow?

THE IDEA · RECURRENCE CFD

Record the flow once, replay it for the temperature, about *260 times* cheaper.

STEP 1

Simulate once

569 min

full CFD: velocity, pressure and temperature

→

STEP 2

Record the motion

600

frames, one per second: where the fluid in each cell goes

→

STEP 3

Replay the heat

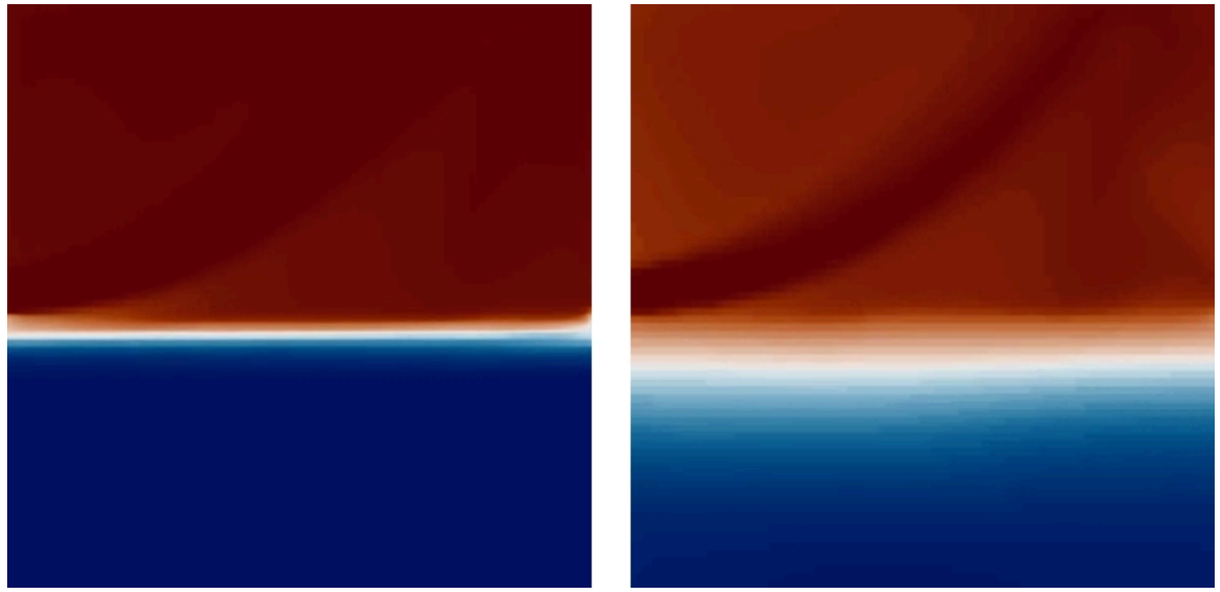
≈ 2 min

no velocity, no pressure: only the temperature is moved

> The catch: the flow is fixed in the recording, so the temperature can no longer change it.

EVIDENCE · THE TEST CASE

Same recorded flow, but the replay's front comes out *four times thicker*.



reference CFD

standard replay, $f_{sd} = 0.5$

mid-plane temperature, 293-333 K, decoded from the rendered frames (mean error ≤ 0.5 K) THESIS CH 5

293 K 313 K 333 K

5 mm

front in the full simulation: a warm layer over cold water

4×

thicker in the standard replay, with the same recorded flow

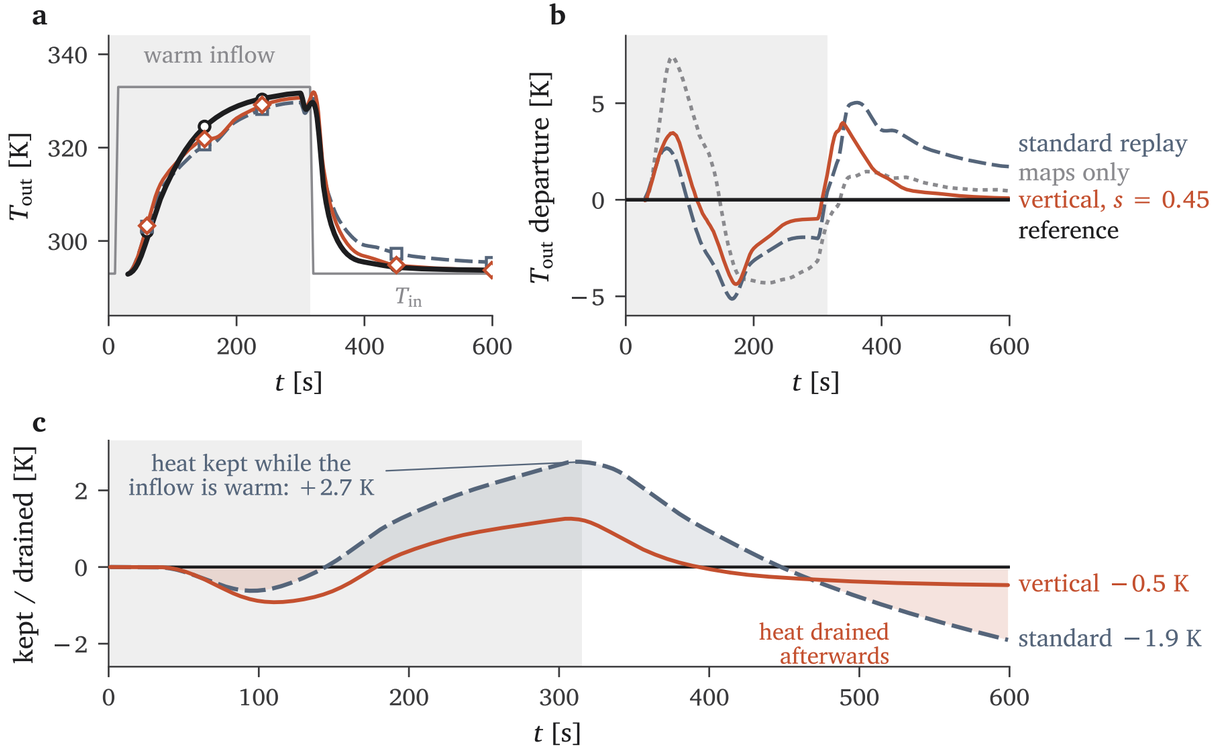
EVIDENCE · THE CAUSE

The outlet sits just below the front, so a smeared front *leaks heat*.

4.6 K

of mean temperature drains through the outlet after the inflow

Found by following the heat: tank budget, then outlet, then one face between two cells. The mixing step smears the front, not the recording.



outlet temperature recovered from the energy balance: heat kept while the inflow is warm, drained afterwards

THESIS FIG 5.10

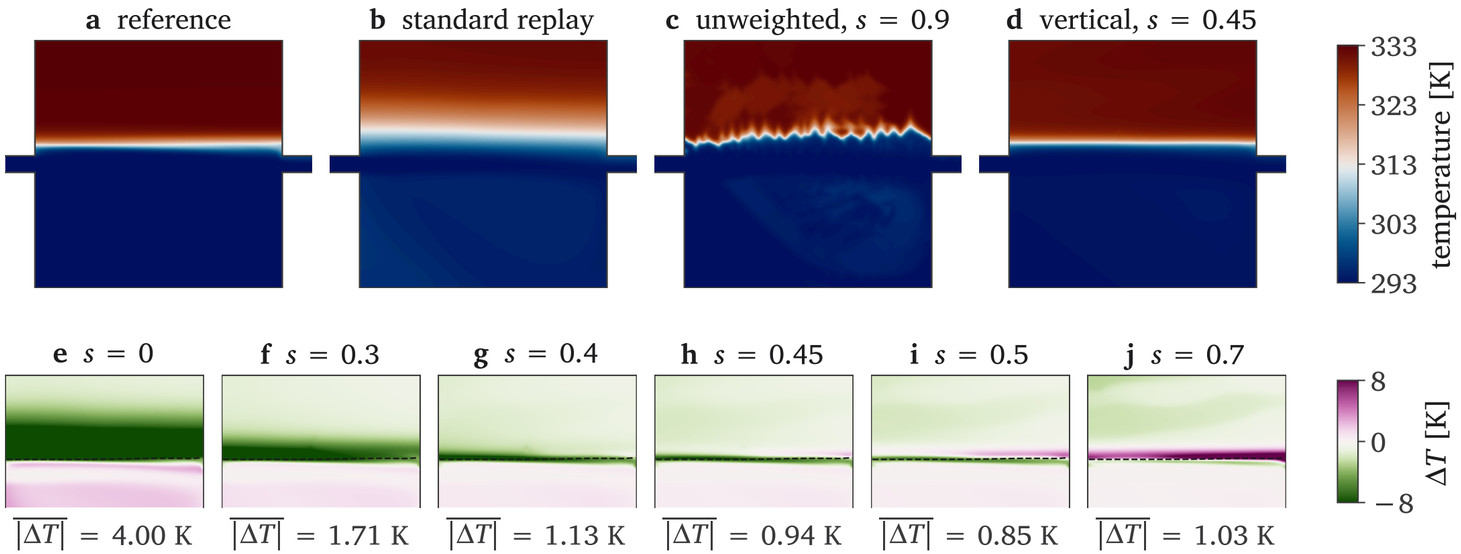
EVIDENCE · THE CORRECTION

The fix moves heat *within physical bounds*, and the front comes back.

15.7 → 5 mm

end front, standard replay and corrected (reference 5.0 mm)

Energy is kept exactly, and no new hot or cold spot can appear.

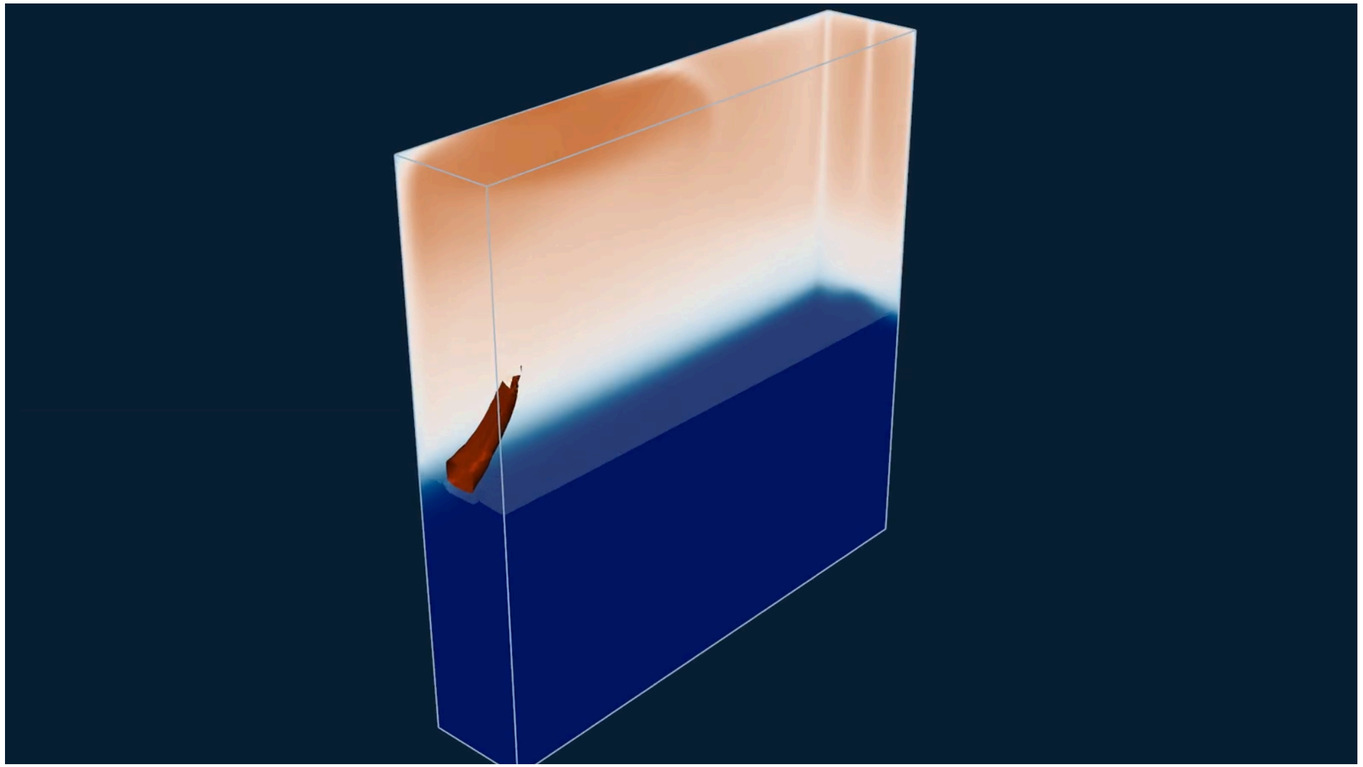


end state and strength ladder; the strength was chosen against the reference

THESIS FIG 5.9

LIMITS · COOLED LID

A recording only serves the *flow regime* it was made in.



cold plumes under a cooled lid: motion that the insulated recording never saw

0.06 K

end error of the heat content with the correction, on 9.62 K of cooling

But the layers separate too slowly while the tank fills. That is missing **transport**, and no local fix can supply it.

NEXT · THREE DIRECTIONS, ALL UNTESTED

Three research directions, each from a *measured weakness*.

UNTESTED

Learn the mixing

instead of a hand-set knob

UNTESTED

Switch by regime

one recording per flow regime

UNTESTED

Physics as tests

energy kept, bounds respected

Before any industrial use: reference data per operating state · recordings that cover the process · checks against real measurements

HOW I WORK

1 Follow the energy until the error has a location.

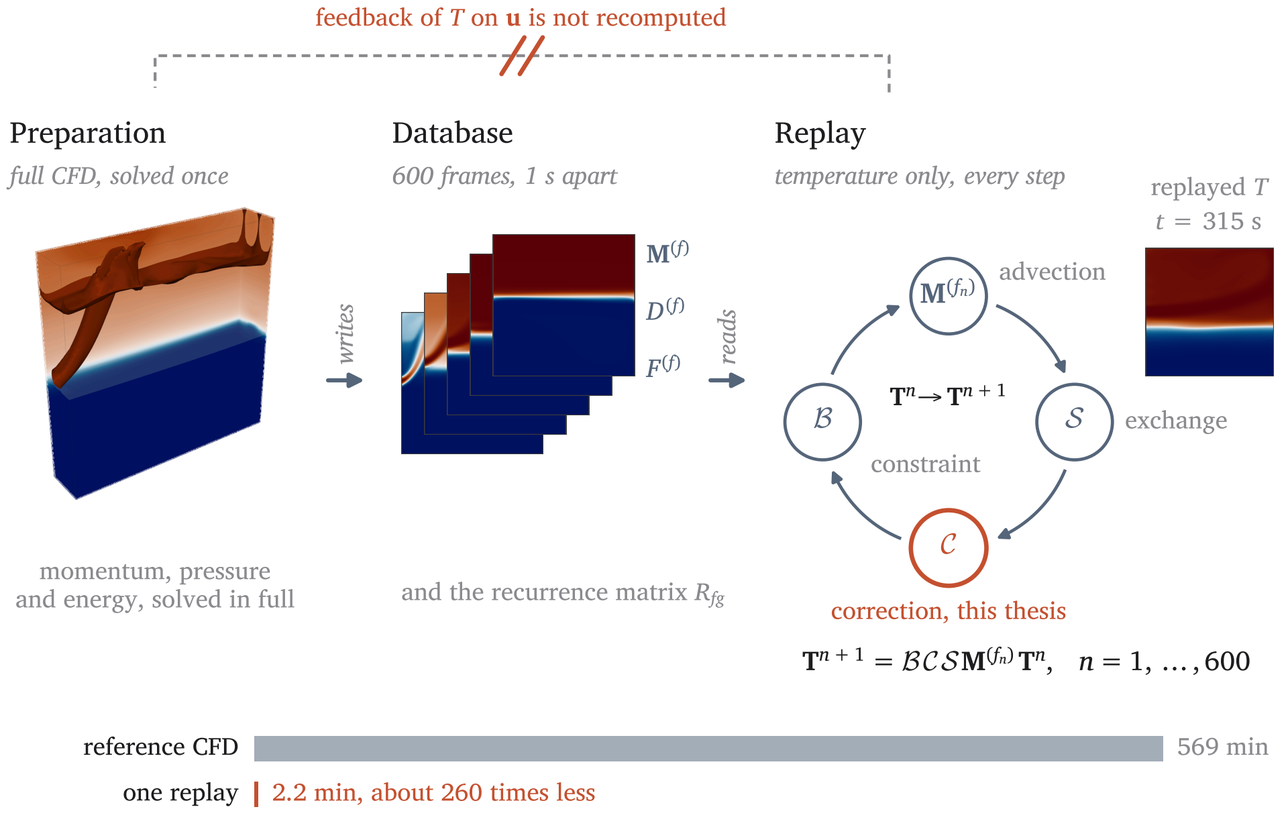
2 Fix it within physical bounds.

3 State where the method stops, with numbers.

Next question: can the replay choose its own correction strength?

BACKUP · METHOD

One replay step: recorded map, face exchange, correction, conservation constraint.



solver time without image output, 569 min against 2.16 min on average; one full CFD run is needed first to record the maps THESIS FIG 2.1

BACKUP · THE INVESTIGATION

From the tank budget, to the outlet, to one face between two cells.

1 · WHOLE TANK

No mixing setting fits both heat and layer height

best heat error 1.07 K with no mixing; best layer height near $f_{sd} = 0.5$; strongest mixing ends 5.08 K too cold

→

2 · OUTLET

A smeared front drains heat

energy balance: 2.7 K kept during the warm inflow, 4.6 K drained after it

→

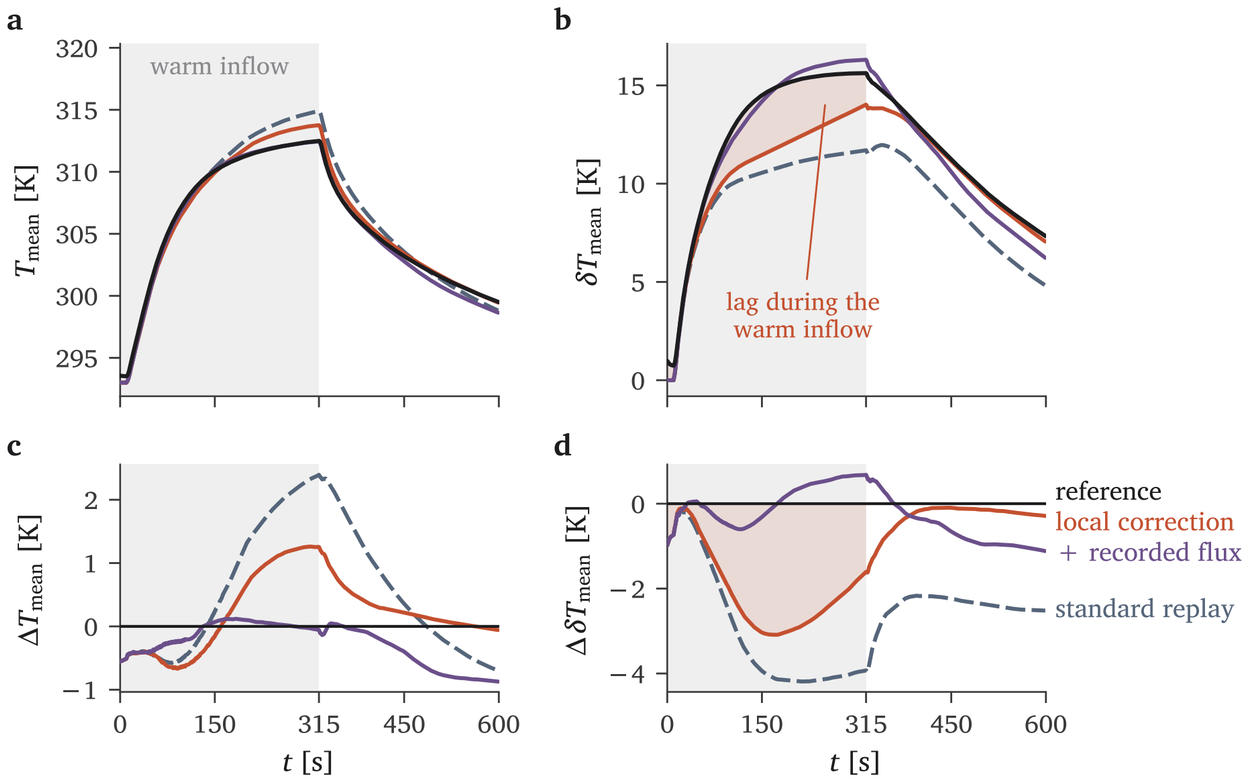
3 · ONE FACE

The mixing step, not the recording

at 240 s: reference 4.5 mm, maps alone 2.5 mm, standard replay 19.3 mm

BACKUP · COOLED LID

Heat right, layering late; a recorded flux helps but overcools.



mean temperature, heterogeneity and their departures

THESIS FIG 6.3

−0.66 → +0.06 K

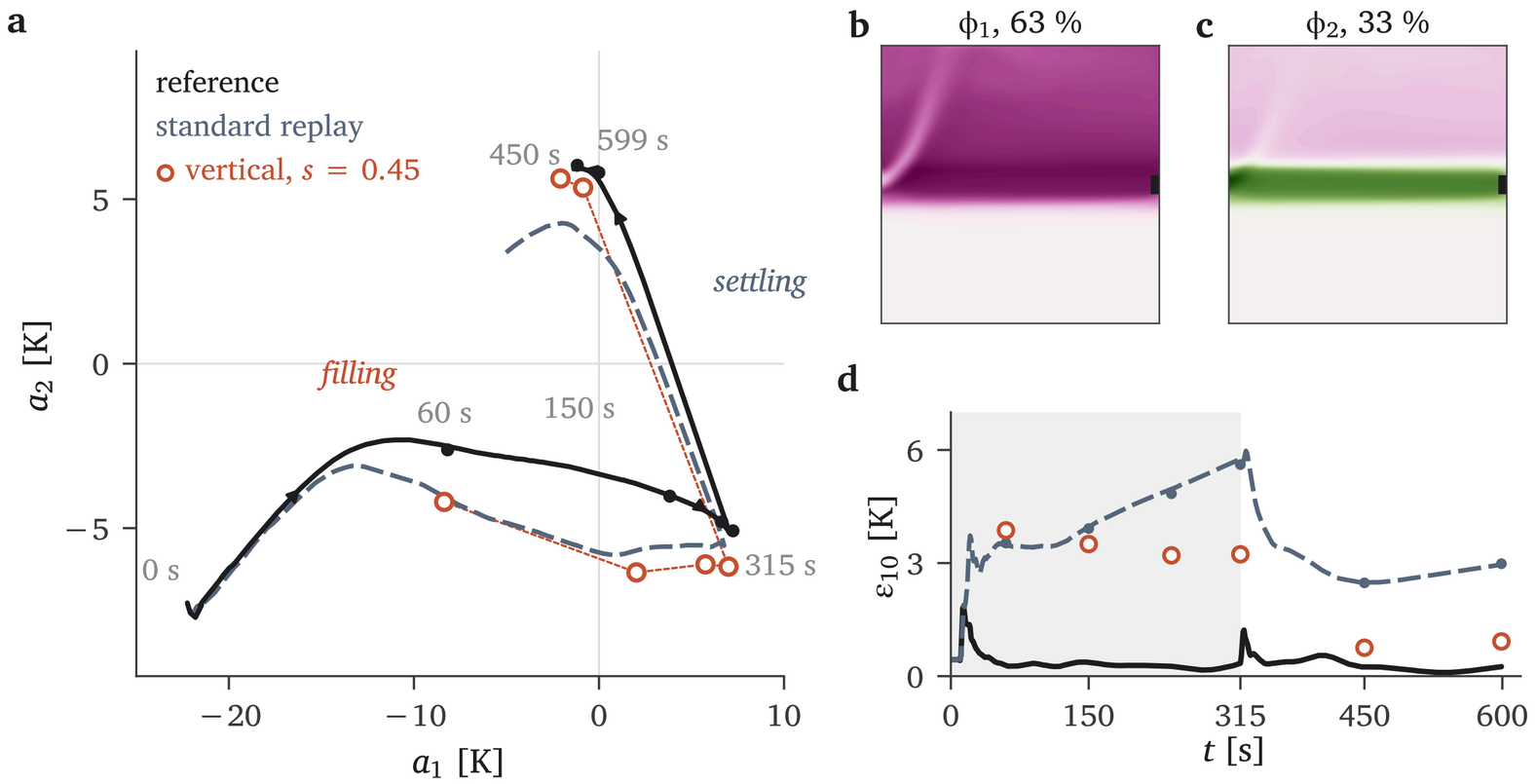
end error on 9.62 K of cooling, standard and corrected ($s \approx 0.55$)

2.25 → 0.45 K

layering lag while filling, without and with a recorded heat flux; the tank then ends 0.86 K too cold

BACKUP · STATE SPACE

Two POD modes hold 96 % of the mid-plane variance of the reference.



runs projected on the modes of the reference; the residual tells how far a field lies from the recorded states

THESIS FIG 5.11

BACKUP · COST

The 260 times is solver time against solver time, without the recording.

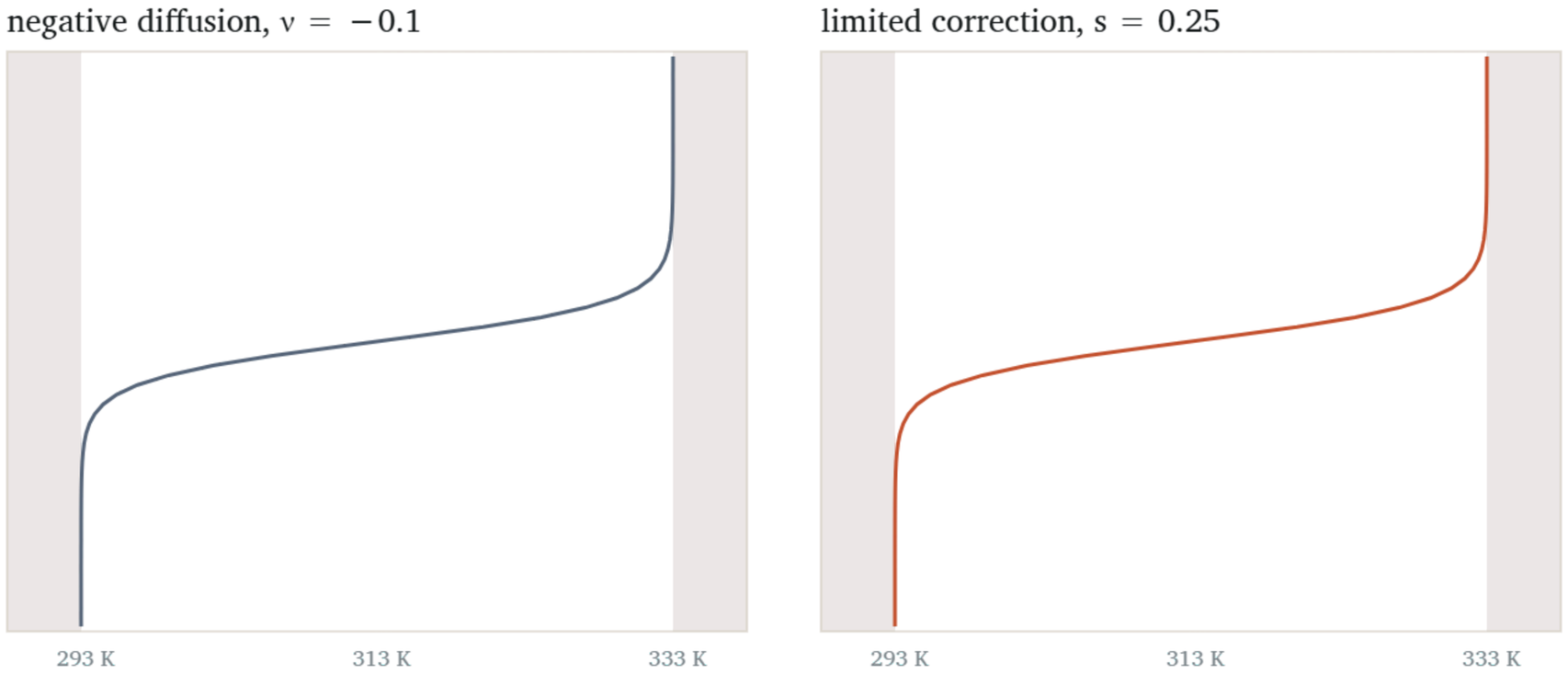
Reference: 569 min for 628 s of flow, image output removed

Replay: 0.81–4.73 min, mean 2.16 min for 600 s; range 120–700x across mixing strengths

The recording is one full CFD run; the saving starts with the second replay

BACKUP · LIMITER

Each face moves at most a share of what both cells can still take.

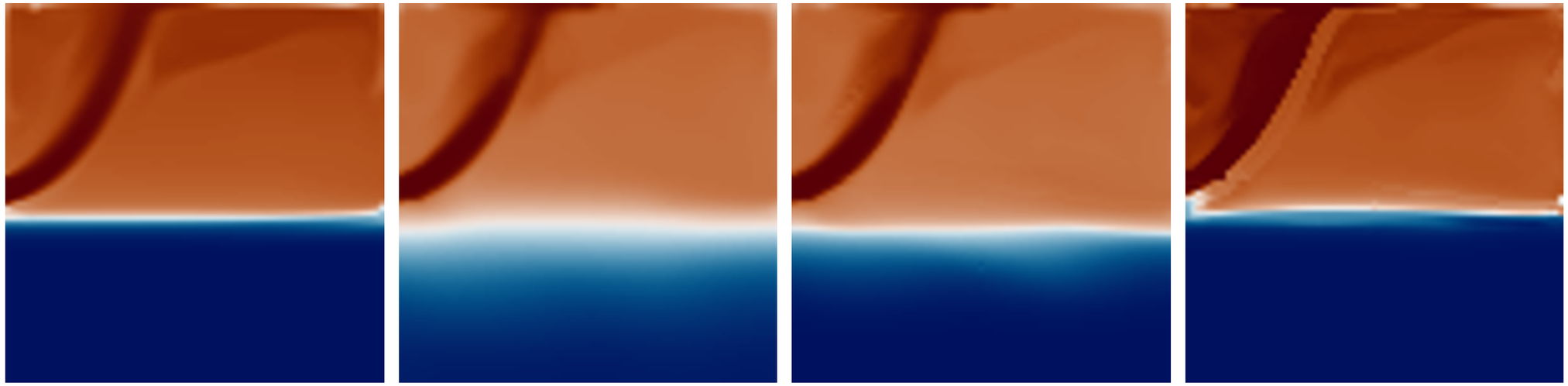


model front: 40 K across 60 cells, ν = −0.1 against limited s = 0.25 (not replay data); shaded: outside the data range

THESIS FIG 4.1

BACKUP · FRAME INSPECTION

Stored snapshots: reference, standard, corrected, and with the recorded flux.



reference CFD | standard replay | corrected, s = 0.475 | + recorded flux

cooled lid, mid-plane temperature 293-333 K

STORED SOLVER SNAPSHOTS

BACKUP · OWN WORK

What I built myself.

The correction stage in the replay library's user hook: limiter, direction weight, stability criterion

A recorded convective-flux channel: written during the recording, applied during the replay

The analysis: energy balances, outlet temperature from the bulk balance, front width, field measures

Every figure from the solver files, with scripts that recompute the numbers

BACKUP · VALIDATION

Before any industrial use, shown on the process's own physics.

- Reference data per operating regime, and what it costs to produce
- Coverage: does the recording contain the motions the process will see?
- Conservation and boundedness of every correction
- Error on the quantity the process cares about, not only on a test tank
- Comparison with measurements (a lab-rig study is running)
- Physics my tank does not have: turbulence, several phases, reaction heat